Erin Kirwin, PhD

email: emkirwin@ualberta.ca pronouns: she/her

Education

[2024] University of Manchester, Manchester, United Kingdom
PhD in Health Economics
Awards: Doctoral Academy Tuition Award
Thesis topic: Methods for the Economic Evaluation of Horizontal Health System Interventions
Supervisor: Matt Sutton, Co-supervisors: Rachel Meacock, Jeff Round

[2014] University of Alberta, Edmonton, Canada
Master of Arts in Economics

[2010] McGill University, Montreal, Canada
Bachelor of Arts (Honors) International Development Studies, Major Economics

Peer-Reviewed Publications

1. **Kirwin E**, Adibnia E, Wiggins M, Sander B, Xie F, Ohinmaa A, Johnson J, Norris C, Rafferty E, MacDonald S, Round J, Canadian Immunization Research Network. *The impact of testing positive versus negative for COVID-19 on health-related quality of life: cross-sectional evidence from the Alberta POST-COVID-19 Follow-up survey*. 2026. Quality of Life Research
2. Wiggins M, Varughese M, Rafferty E, van Katwyck S, McCabe C, Round J, **Kirwin E.** *The Decision Uncertainty Toolkit: Risk Measures and Visual Outputs to Support Decision Making During Public Health Crises*. 2025. PLOS ONE
3. Wiggins M, Round J, **Kirwin E.** *The cost-effectiveness of a diabetes prevention program in Alberta, Canada*. 2025. International Journal of Technology Assessment in Health Care.
4. Round J, **Kirwin E**, van Katwyck S, McCabe C. *Improving Transparency of Decision Models Through the Application of Decision Analytic Models with Omitted Objects Displayed (DAMWOOD)*. 2024. PharmacoEconomics.
5. Rafferty E, Unsal A, **Kirwin E.** *Healthcare Costs and Effects of Post-COVID-19 Condition in Canada*. 2023. Canada Communicable Disease Report.
6. Akpinar I, **Kirwin E**, Chojecki D, Tjosvold L, Round J. *A systematic review of the accessibility, acceptability, safety, efficiency, clinical effectiveness and cost-effectiveness of private providers of elective surgical services compared with public providers*. 2023. International Journal of Health Technology Assessment.
7. **Kirwin E**, Meacock R, Round J, Sutton M. *The Diagonal Approach: A theoretic framework for the economic evaluation of vertical and horizontal interventions in healthcare*. 2022. Social Science & Medicine.
8. **Kirwin E**, Bond K, Round J, McCabe C. *A Conceptual Framework for Life-Cycle Health Technology Assessment*. 2022. Value in Health [Editor's Choice]
9. **Kirwin E**, Rafferty E, Harback K, Round J, McCabe C. *A Net Benefit Approach for the Optimal Allocation of a COVID-19 Vaccine*. 2021. PharmacoEconomics.
10. Mulberry N, Tupper P, **Kirwin E**, McCabe C, Colijn C. *Vaccine rollout strategies: The case for vaccinating essential workers early*. 2021. PLOS Global Public Health.
11. Briggs A, Goldstein D, **Kirwin E**, Meacock R, Pandya A, Vanness D, Wisloff T. *Estimating (quality-adjusted) life-year losses associated with deaths: with application to COVID-19*. 2020. Health Economics.

12. **Kirwin E**, Varughese M, Waldner D, Simmonds K, Joffe A, Smith S. *Comparing methods to estimate incremental inpatient costs and length of stay due to methicillin-resistant Staphylococcus aureus in Alberta, Canada*. 2019. BMC Health Services Research.
13. Barichello S, Nguyen T, Lu D, Ismond K, Loomes D, **Kirwin E**, Wang H, Chang D, Svenson L, Fedorak R. *Comparative Effectiveness and Cost-Effectiveness Analysis of a Urine Test versus Alternative Colorectal Cancer Screening Strategies*. 2018. American Journal of Gastroenterology.
14. Beilman C, **Kirwin E**, Ma C, McCabe C, Fedorak R, Halloran B. *Early Initiation of Anti-TNF Therapy is Cost-Saving Compared to Late Initiation for Patients with Crohn's Disease*. 2017. Clinical Gastroenterology & Hepatology.
15. Dover D, **Kirwin E**, Hernandez N, Nelson K. *Pandemic Risk Assessment Model* (PRAM): *A Mathematical Modeling Approach to Pandemic Influenza Planning*. 2016. Epidemiology and Infection.

Working Papers

16. Wiggins M, Locke N, **Kirwin E,** Round J. *Development and feasibility of a novel discrete choice experiment for the joint elicitation of child and caregiver health-related quality of life*. Under review.
17. Barati E, **Kirwin E.** *Indirect and Direct Non-Medical Costs in Patients with Type-2 Diabetes Mellitus: A Meta-Analysis*
18. **Kirwin E,** Rafferty E**,** MacDonald S. *Pharmacist, physician, or public health? How individual-level factors impact influenza immunization provider choice and subsequent coverage in Alberta, Canada.*
19. **Kirwin E**, Meacock R, Round J, Sutton M. *The impact of diagnostic imaging wait time on patient mortality and health system cost*.
20. **Kirwin E**, Paulden M, McCabe C, Round J, Sutton M, Meacock R. *The risk-based price: incorporating uncertainty and risk attitudes in health technology pricing*. [Pre-Print on SSRN]
21. **Kirwin E**, Meacock R, Round J, Sutton M. *Costs and Benefits of Incentivizing Physicians to Enhance Care of Patients with Complex Conditions: Econometric Evidence from Patient Migration*.

Other Published Work

22. **Kirwin E**, Bond K, Round J, McCabe C. *Response to: Comments on A Conceptual Framework for Life-Cycle Health Technology Assessment*. 2023. Value in Health.
23. **Kirwin E,** MacDonald S, Simmonds K. *Profiles in Epidemiology: Dr. Larry Svenson*. 2022. American Journal of Epidemiology.
24. **Kirwin E.** *Optimizing MRI Service Capacity Phase 2: The impact of MRI wait time on patient mortality, hospital length of stay, and health system cost*. 2025. Institute of Health Economics.
25. Wiggins M, **Kirwin E**. *MRI Capacity Phase 3: Optimizing MRI wait times by scan urgency to improve health system outcomes*. 2025. Institute of Health Economics.

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Appointments

[10 2025 – Present] Institute of Health Economics **Managing Principal Health Economist**

[07 2025 – Present] University of Alberta **Adjunct Professor**

[03 2024 – Present] University of Manchester **Honorary Senior Research Fellow**

[07 2019 – 10 2025] Institute of Health Economics **Senior Principal Health Economist**

[07 2017 – 07 2019] Alberta Health **Manager, Advanced Analytics**

[01 2011 – 07 2017] Alberta Health **Public Health Economist**

Research Funding

Co-Primary Investigator. *The Decision Uncertainty Toolkit: Estimating and communicating decision uncertainty within infectious disease models*. Canadian Network for Modelling Infectious Diseases. ($70,000 from 2021-2022)

Collaborator. *One-society Network: Mathematical modelling of multi-sectoral impacts of pandemics and control policies*. Natural Sciences and Engineering Research Council (NSERC) Emerging Infectious Disease Modelling Initiative. ($1,250,000 from 2021-2023) *I lead the grant application development under PI Dr. Chris McCabe

Co-Investigator. *Estimation of long-term COVID-19 health state utility values. Canadian Institutes for Health Research*. Vaccine Readiness Grant: Modelling and Economics Research Network. ($115,165 from 2021-2022)

Trainee. *Canadian Immunization Research Network (CIRN). Public Health Agency of Canada*. Canadian Institutes for Health Research. ($6,600,000 from 2014-2017; $10,083,000 from 2017-2022)

Additional Funding

Conference support. **University of Manchester Doctoral Academy Conference Support Fund.** Society for Medical Decision Making Annual Meeting, Philadelphia, USA. (October 2023, £500)

Conference support. **HTAi.** HTAi Annual Meeting, Adelaide, Australia. (June 2023, $2,800 USD)

Conference support. **University of Manchester Doctoral Academy Conference Support Fund.** European Health Economics Association (EuHEA) PhD and Early Career Research Meeting, Galway, Ireland. (August 2022, £500)

Conference support. **Health Economics Study Group.** Health Economics Study Group Meeting Summer 2022, Sheffield UK. (June 2022, approx. £400)

Travel Support. **Quebec Youth Travel Forum.** World Youth Congress, Canadian Delegate, Istanbul, Turkey. (June 2010, $600 CAD)

Stipend. **Burgess Family Award, McGill Arts Internship Office.** Field Assistant Intern, Hatua Kenya, Kisumu, Kenya. (Summer 2009, $3000 CAD)

Teaching: Sessional and Guest Lectures

University of York Centre for Health Economics. Ongoing
University of Alberta Department of Mathematics. March 2023, November 2025
University of Alberta Department of Pharmaceutical Sciences, November 2025
McGill University Quantitative Life Sciences. November 2023

Conference Course Instruction

Kirwin E, van Katwyck S, McCabe C, Round J. *Research Oriented Market Access for High-Cost Therapies, Principles and Practice*. 2022, 2023. Health Technology Assessment international (HTAi) Conference.

Conference Papers and Presentations (Posters Excluded)

Kirwin E. *Panelist: Medical Imaging Capacity*. 2026. Canada's Drug Agency Symposium [Upcoming]

Kirwin E, Meacock R, Round J, Sutton M. *Costs and Benefits of Incentivizing Physicians to Enhance Care of Patients with Complex Conditions: Econometric Evidence from Patient Migration*. 2025. Vancouver Health Economics Methods Meeting [Paper discussed in dedicated session]

Kirwin E, Paulden M, McCabe C, Round J, Sutton M, Meacock R. *The Risk-Based Price: incorporating uncertainty and risk attitudes in health technology pricing*. 2023. Vancouver Health Economics Methods Meeting [Paper discussed in dedicated session]

Kirwin E, McCabe C, Round J. *The Decision Uncertainty Toolkit: Risk Measures and Visual Outputs to Support Health Technology Decision-making During Public Health Crises*. 2023. HTAi Annual Meeting

Kirwin E, Paulden M, McCabe C, Round J, Sutton M, Meacock R. *The Risk-Based Price*. 2023. HTAi Annual Meeting

Kirwin E, Meacock R, Round J, Sutton M. *Costs and Benefits of Incentivizing Physicians to Enhance Care of Patients with Complex Conditions: Econometric Evidence from Patient Migration*. 2022. European Health Economics Association PhD Student and Early Career Researcher Conference, Ireland. [Paper discussed in dedicated session]

Kirwin E, Meacock R, Round J, Sutton M. *Costs and Benefits of Incentivizing Physicians to Enhance Care of Patients with Complex Conditions: Econometric Evidence from Patient Migration*. 2022. Sheffield Health Economics Study Group. [Paper discussed in dedicated session]

Round J, **Kirwin E**, van Katwyck S, McCabe C. *The Morals of Models: Improving Public Understanding of Decision Models Through the Application of Decision Analytic Models With Omitted Objects Displayed (DAMWOOD)*. Sheffield Health Economics Study Group. [Paper discussed in dedicated session]

Kirwin E, Paulden M, McCabe C, Round J, Sutton M, Meacock R. *The Risk-Based Price*. 2021. CADTH [Oral presentation]

Kirwin E, Paulden M, McCabe C, Round J, Sutton M, Meacock R. *The Risk-Based Price*. 2021. Cambridge Health Economics Study Group. [Paper discussed in dedicated session]

Kirwin E, Hopkin G, Bond K, Round J, McCabe C. *Health Technology Assessment Methods for Research Oriented Market Access*. 2020. Society for Medical Decision Making North American Meeting. [Oral hybrid presentation]

Invited Oral Presentations (Selected)

Kirwin E. 2023. University of York Centre for Health Economics. *The Risk-Based Price: incorporating uncertainty and risk attitudes in health technology pricing*.

Kirwin E. 2022. University of Alberta Department of Economics. *The Diagonal Approach: A Theoretic Framework for the Economic Evaluation of Vertical and Horizontal Interventions in Healthcare*.

Kirwin E. 2021. Pennsylvania State University Health Services Research Colloquium. *The Diagonal Approach*.

Kirwin E. 2021. Pennsylvania State University, Seminar to the Medical School at Hershey. *The Risk-Based Price: incorporating uncertainty and risk attitudes in health technology pricing*.

Kirwin E. 2021. NOAHE Rounds V. *The Diagonal Approach: A Theoretic Framework for the Economic Evaluation of Vertical and Horizontal Interventions in Healthcare*.

Kirwin E. 2021. The United Kingdom Office of Health Economics. *The Risk-Based Price: incorporating uncertainty and risk attitudes in health technology pricing*.

Kirwin E. 2020. Alberta Health Economics Study Group. *Real Time Epidemiological Modelling to inform the COVID-19 Pandemic Response*. **Keynote**.

Kirwin E. 2019. NOAHE Rounds III. *Administrative Data Analytics at Alberta Health*.

Kirwin E 2013- 2018. Alberta Advisory Committee on Immunization. *Various evidence and modelling presentations*.

Kirwin E. 2016. Canadian Chronic Disease Surveillance System Economic Committee National Meeting. *Using Economic and Epidemiologic Data for Chronic Disease Surveillance*.

Kirwin E. 2014. Western Canada Immunization Forum. *HPV Vaccine for Boys: A Cost Effectiveness Analysis*.

Software

R, SAS, Mathematica, Stata

Conference/Meeting Organization

Alberta Health Economics Study Group, 2022, 2025, 2026

Memberships

Network of Alberta Health Economists: Member, **Current**. Network manager.

UK Health Economics Study Group. Member, **Current**

Professional Society for Health Economics and Outcomes Research: Member, **Current**

Society for Medical Decision Making: Member, **Current**

Canadian Chronic Disease Surveillance System Economic Committee: Alberta Delegate

Alberta PROMs and EQ-5D Research and Support Unit: Scientific Advisory Committee Member

Additional Training

[2020] Michigan State University, Professor Jeff Wooldridge, Virtual
ÊSTIMATE: The Reduced Form
Econometrics course with a focus on understanding when various models and estimation methods are appropriate as well as how to conduct proper inference in a variety of settings.

[2015] Health Economics Research Unit (HERU), University of Aberdeen
and O'Brien Institute for Public Health, University of Calgary, Banff, Canada
Using Discrete Choice Experiments in Health Economics: Theoretical and Practical Issues
Introduction to discrete choice experiments with hands-on experience of the design, questionnaire development, data input, analysis and interpretation.

[2014] SAS Institute, Virtual
Programming I, II, Macro Language
Programming Essentials, Data Management Techniques, Macro Language

[2012] PIMS-Mprime-CDM Summer School, University of Alberta, Edmonton, Canada
Mathematical Modeling of Infectious Diseases
Theory and methods for modeling infectious diseases with MATLAB and Mathematica.

Journal Refereeing

Health Economics, PharmacoEconomics, PLOS One, International Journal of Health Technology Assessment